Lecture 6
Building a Reliable Dataset for a RAG-Based Chatbot

From authoritative documents to an evaluation-ready knowledge base



ITC-C508 – Deep Learning (NLP)
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Learning Outcomes

By the end of the lecture, students should be able to:

- Distinguish the knowledge base from the independent test set.
- Define a clear chatbot scope and document inclusion/exclusion rules.
- Collect, inspect, clean, and version authoritative documents.
- Create chunks with meaningful boundaries and traceable metadata.
- Prepare answerable, unanswerable, and out-of-scope test questions.
- Record reference answers and supporting evidence.
- Prevent data leakage and design fair RAG experiments.
- Submit a reproducible dataset preparation record.

Required output

A documented knowledge base + an independent, evidence-backed test set.

The Two Components of a RAG Dataset

A. Knowledge Base

The collection of source documents that the chatbot can retrieve from.

Examples:
- Official policies
- Manuals and FAQs
- Research or technical documents
- Approved institutional webpages

B. Test-Question Set

A separate collection of questions used to evaluate the chatbot.

Each record should include:
- Question
- Expected answer
- Supporting passage
- Difficulty and scope labels

Core rule:

The knowledge base supplies evidence; the test set measures whether the system can use that evidence correctly.

Why Dataset Reliability Matters

1. Source problem

Incorrect, outdated, duplicated, or unauthorized content enters the knowledge base.

2. Retrieval problem

The correct information exists, but the retriever fails to return the relevant chunk.

3. Generation problem

The LLM receives weak or ambiguous evidence and produces an unsupported answer.

Dataset preparation directly addresses source quality and makes retrieval/generation failures easier to diagnose.

Characteristics of a Reliable Knowledge Base

Relevant
Supports the chatbot's stated purpose.

Authoritative
Comes from a trustworthy or approved source.

Current
Uses the correct version or effective date.

Complete
Covers important topics without major gaps.

Traceable
Source, processing, and version can be audited.

Retrievable
Can be chunked and retrieved without losing meaning.

Quality-control question

Can another researcher understand where each chunk came from and why it belongs?

Step 1 — Define the Chatbot's Scope

In scope

Questions the chatbot is designed to answer.

Example:
- Enrollment requirements
- Application deadlines
- Required documents
- Official procedures

Out of scope

Questions outside the knowledge domain or beyond the chatbot's authority.

Example:
- Personalized legal advice
- Medical diagnosis
- Unverified rumors
- Information not supported by sources

Student task
Write one scope statement, five intended questions, and three questions the chatbot should redirect.

Step 2 — Collect Authorized Documents

1. **Identify** — Find the responsible office, institution, publisher, or authority.
2. **Verify** — Check whether the source is official, current, and appropriate.
3. **Record** — Save title, URL/file path, date, version, and access date.
4. **Preserve** — Keep the original file before cleaning or transformation.

Important
Publicly accessible does not automatically mean authorized for unrestricted reuse.

Step 3 — Create a Document Inventory

Doc ID	Title	Version / Date	Languag	Status
D001	Enrollment Policy	v3 • 2026-01	English	Approved
D002	Application Guide	2026 edition	English	Approved
D003	FAQ Webpage	Accessed 2026-09	English	Review
D004	Student Handbook	v5 • 2025-08	English	Approved

Add these fields

- Source organization
- Original URL/path
- Access date
- Authorization notes
- Processing status
- Checksum (optional)
- Relevance rationale

Additional fields to record: Source organization, original URL or path, access date, authorization notes, processing status, optional checksum, and relevance rationale.

Why it matters
The inventory becomes the audit trail for the knowledge base.

Step 4 — Clean and Inspect the Documents

Before cleaning

- Repeated headers and footers
- Page numbers mixed into sentences
- OCR errors and broken words
- Duplicate paragraphs
- Missing section boundaries
- Tables extracted incorrectly

After cleaning

- Meaning preserved
- Headings and sections retained
- Tables checked manually
- Duplicates removed or flagged
- OCR corrections recorded
- Original file retained

Rule: Cleaning changes formatting and noise—not the meaning of the source.

Step 5 — Split Documents into Meaningful Chunks

Chunking should preserve enough context for a passage to answer a question.

Original document

Enrollment Policy

1. Eligibility
2. Required documents
3. Deadlines
4. Application procedure

Chunk 1

Eligibility + context

Chunk 2

Required documents

Chunk 3

Deadlines + procedure

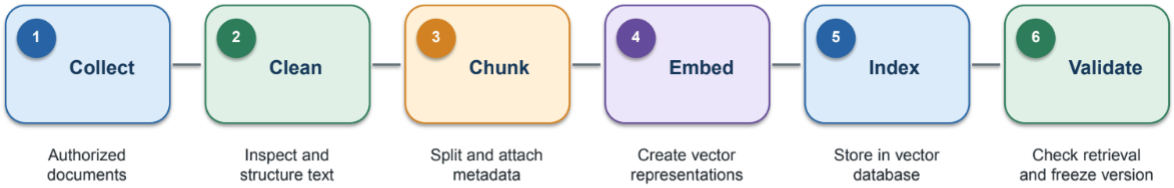
Record chunk metadata

- Chunk ID
- Document ID
- Section / heading
- Page number
- Character/token count
- Chunking strategy
- Overlap setting

Experiment variable

Chunk size, overlap, and boundary strategy should be tested—not assumed to be universally optimal.

Knowledge-Base Preparation Pipeline



Freeze a version

Once the evaluation begins, record the exact documents, chunks, embeddings, index, and configuration used.

Step 6 — Design an Independent Test Set

The test set should represent realistic user questions—not only questions that are easy to answer.

Direct factual

Answer appears in one passage.

Multi-step

Requires combining facts or sections.

Difficult

Requires comparison, multi-hop retrieval, or interpretation.

Unanswerable

Topic is relevant, but evidence is absent.

Out of scope

Question is outside the chatbot's purpose.

Minimum record

Question ID • question • expected answer • evidence • difficulty • answerability • scope

Test Questions: Difficulty Levels

Easy

One passage directly states the answer.

Example question

What documents are required?

Moderate

Requires combining facts or interpreting a section.

Example question

What is required before submitting the application?

Difficult

Requires multi-hop retrieval, comparison, or reasoning.

Example question

Compare requirements for two applicant categories.

Operational definition

Difficulty should be assigned using explicit criteria, not personal intuition alone.

Test Questions: Scope and Answerability

In scope + answerable

The question belongs to the domain and the knowledge base contains supporting evidence.

Expected behavior:
Answer with evidence.

In scope + unanswerable

The question belongs to the domain, but the documents do not contain enough evidence.

Expected behavior:
State that the information is unavailable.

Out of scope

The question is outside the chatbot's intended purpose.

Expected behavior:
Decline or redirect appropriately.

Step 7 — Prepare Reference Answers and Evidence

Question	Reference answer	Evidence	Scoring criterion
What documents are required?	Valid ID and completed form.	D002, p. 4, §2	All required items included
When is the deadline?	30 September 2026.	D001, p. 2, §1	Correct date and context
Can late applications be accepted?	Only under stated exceptions.	D001, p. 5, §4	Mentions the condition

Important distinction

A reference answer summarizes what is expected; evidence identifies where that answer is supported.

Data Leakage: A Threat to Research Validity

Leakage occurs when evaluation information influences the system before final testing.

1. Create test question

A student writes a question and expected answer.

2. Tune using test

The student notices failure and adds a special chunk containing the answer.

3. Inflated result

The final score is higher because the test influenced the knowledge base.

Research principle

The final test must remain independent of tuning decisions.

How to Prevent Data Leakage

1. Define scope and collection rules
2. Prepare and freeze the initial test set
3. Split development and evaluation questions
4. Tune using development data only
5. Lock the final configuration
6. Evaluate once on the untouched test set
7. Record later changes as new experiments

For example:

Development question	Evaluation question
D06: What steps must a student complete before submitting the application?	E01: What documents must an applicant submit, and what is the purpose of each?
D09: How do the application and enrollment procedures differ?	E02: Compare the enrollment requirements for two applicant categories.
D11: Are late applications accepted?	E03: If an applicant misses the deadline, what options are explicitly allowed?

Dataset Reliability and Fair Experiments

Control / keep fixed

- Test questions
- Evaluation rubric
- Evaluation procedure
- Target task and scope
- Reporting format

Change / compare

- Chunk size
- Chunk overlap
- Embedding model
- Retriever or reranker
- Prompt configuration

Measure / report

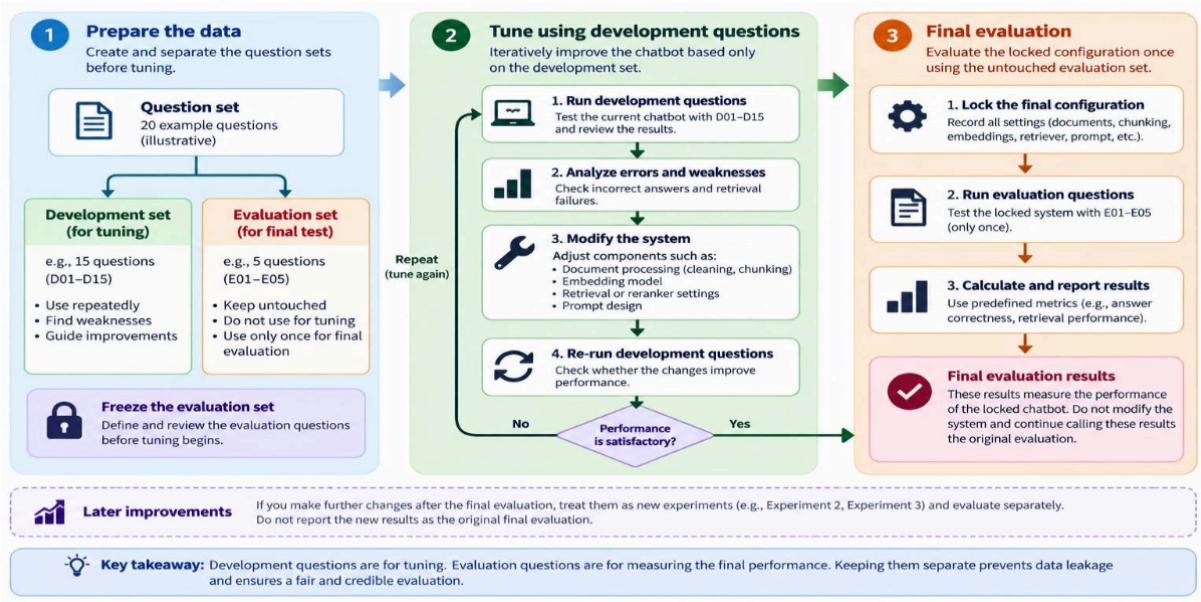
- Recall@k
- Precision@k, if applicable
- Answer correctness
- Faithfulness / groundedness
- Failure examples

Fair comparison

Change one major factor at a time, or justify a design that studies interactions.

Leakage-Controlled Tuning of a RAG Chatbot

Use development questions to improve the system; reserve evaluation questions for the final, independent assessment.



What Students Should Document

Deliverable A — Knowledge Base

- Document inventory and provenance
- Original files and versions
- Cleaning/transformation log
- Chunking strategy and parameters
- Chunk metadata and examples
- Embedding and vector-store settings
- Final knowledge-base version

Deliverable B — Test Set

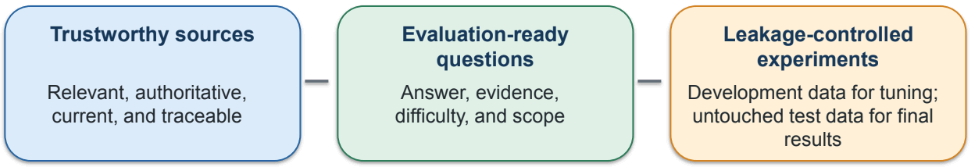
- Independent questions
- Difficulty and answerability labels
- Expected/reference answers
- Supporting passages and source IDs
- Evaluation rubric
- Development/evaluation split
- Leakage-prevention record

Assignment

Prepare at least 20 manually authored test questions for your RAG research topic.

Key Takeaway

A reliable RAG dataset connects trustworthy knowledge, carefully designed questions, and strict separation between development and evaluation.



Research consequence

Reliable data makes experiments more defensible; it does not guarantee that every chatbot answer is correct.